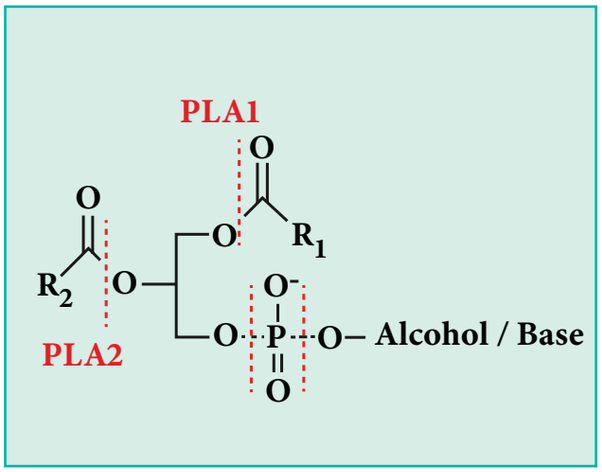
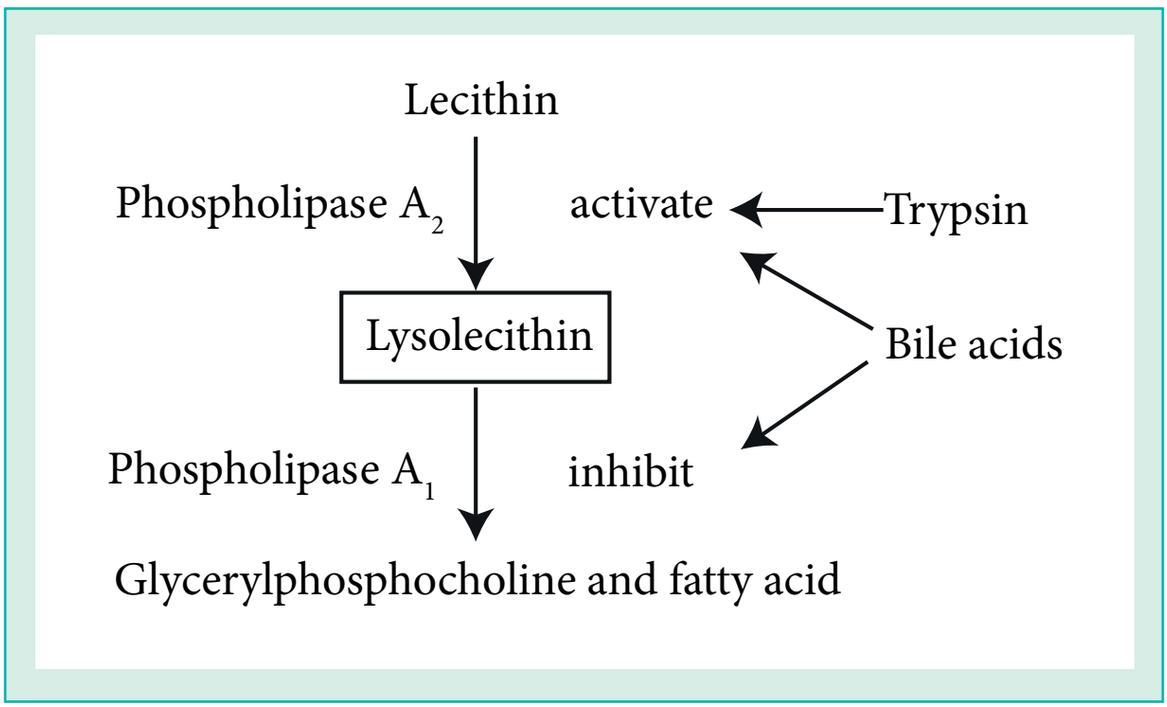
5.5.4 Degradation of Phospholipids

Phospholipases are enzymes that degrade phospholipids through hydrolytic cleavage of carboxy- and phospho-diester bonds. The enzymes are classified as phospholipases A1, A2, C or D depending on the site of hydrolysis at the sn-1 or sn-2 acyl ester bond, at the glycerol phosphate bond or at the glycerol phosphate-base phosphodiester bond, respectively.



5.5.5 Lysolecithins formation

Lysophosphatidylcholines (LPC, lysoPC), also called lysolecithins, are a class of chemical compounds which are derived from phosphatidylcholines. They are formed in the duodenum by the partial hydrolysis of lecithin in bile by phospholipase A_2 of pancreatic juice, which removes one of the terminal fatty acid groups. This reaction is activated by acids and trypsin. Lysolecithin may be further hydrolyzed to glyceryl phosphorylcholine by phospholipase A_1, but this reaction is inhibited by bile acids.



5.5.6 Effects of lysolecithin

Lysolecithin is highly toxic to cell membranes and it is a strong detergent similar to bile salts. It can demyelinate nerves and destroy red blood cells. Due to its emulsification properties, it helps to make the consistency of products smooth and easy to spread. Because of this, LPC is widely used in food and pharmaceutical industry and also in livestock and poultry feeds.

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